👩‍🏫 Staff Guide · Horticulture 1 · Unit 3 Parts of the Plant (Level 2)

Level 2 = about 3rd-grade reading. SLE students stay in the class teacher's lesson and use these pieces next to it: same topics, same order, same points.

1. How the pieces line up

Gen ed piece (Canvas: "Unit 3 Parts of the Plant (textbook Unit 3): leaves, stems, roots, flowers worksheets, leaf collection, growth habits, photosynthesis, pollination (Silence of the Bees), Unit 3 quiz (28 Qs)")	**Level 2 piece**
Textbook Unit 3 + teacher notes (stems, roots)	Slides + notes A (facts rewritten; textbook scans are not posted)
Leaves worksheet + leaf collection	Leaf Collection Walk lab packet + notes B
Photosynthesis	Slides + notes C
Flowers worksheet + flower dissection	Flower Dissection lab packet
Silence of the Bees + worksheet	Pollination slides + video guide
Unit 3 quiz (28 Qs)	Modified quiz
Class slides	Deck: 19 slides, Words to Know first, hold-up-1-2-3 checks, recap
Her worksheets (grouped into lessons)	Guided notes: 14 items + key
Class quiz / notes check	Quiz: 10 questions + key
Vocabulary	9 picture cards (cut apart)
—	1 game on the class site: 10 questions + a matching round

2. Supports built in

- Short lines, key words bold, word bank in every notes box
- Quiz: 3 choices, no "all of the above"
- Picture anchors for root/stem/leaf/flower
- Real plant to touch: point to each part

3. Add these as the IEP allows

- Read-aloud · extended time · separate setting
- Printed slides · videos paused at the "look for" part

4. What to grade

- Unit 3 quiz: Level 1 = 8 picture items, Level 2 = 10 items; scale to class points
- **Notes**: same points as her worksheets for that lesson when complete.
- **Quiz**: score, then scale to the class points.
- These are **modifications** (less content). The IEP must allow modified grading. Check with the case manager.

5. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm in her files say "check with the class teacher".

🚧 Under construction (check with the class teacher)

- **Pollination mini videos**: This page is empty in the export.
- **Silence of the Bees**: Full PBS episode upload on YouTube; staff should use a short official PBS clip.

💗 Note

Bee safety: check health plans for bee-sting allergies before outdoor or greenhouse work near flowers.

⚠️ Safety: tools and plants

- **Tools**: students may use all tools (pruners, knives, floral wire, low-temp glue guns) **with an adult right there**. Staff hand tools out, stay at the station, and count them back in. Cut-resistant gloves if possible.
- **Poinsettia sap** is milky and can irritate skin and eyes. Gloves on, no rubbing eyes, wash hands after.
- **Latex**: poinsettias are in the latex family. Students with a latex allergy wear non-latex (nitrile) gloves or skip handling; check health plans.
- **Bees and pollen**: check IEP/health plans for bee-sting or pollen allergies and know where the EpiPen is before greenhouse work.
- **Mouths**: no plants, soil, seeds, berries, fertilizer or rooting hormone in mouths. Students who mouth objects or have pica need one-to-one eyes in the greenhouse. Many plants (holly and mistletoe berries, some houseplants) are poisonous.
- **Chemicals**: rooting hormone and fertilizer with gloves, then wash hands. Pesticides and growth regulators are staff/teacher only; students never spray or handle them.
- **Greenhouse**: wet floors and hoses (slips), heat (water breaks), wet dry potting mix first (dust).

👩🏾‍🏫 Staff Guide · Flower Dissection (Level 2)

Horticulture 1 · Unit 3 · goes with the class “Flowers Worksheet / flower dissection” (4 pts). Level 2 = about 3rd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Task card**: Modified from the class “Flowers Worksheet / flower dissection”. 8 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 3 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- Show me the first step.
- Which tool do we need now?
- Where do our hands go when the tool is cutting?
- What should we do when we finish?

3. Modifications

- Task card instead of written directions
- Sentence starters on the reflection
- Partner reads directions aloud

4. What to grade (4 pts)

- Steps done (photo proof)
- Reflection: 2 sentences per box
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver (SLE).

⚠️ Safety: tools and plants

- **Tools**: students may use all tools (pruners, knives, floral wire, low-temp glue guns) **with an adult right there**. Staff hand tools out, stay at the station, and count them back in. Cut-resistant gloves if possible.
- **Poinsettia sap** is milky and can irritate skin and eyes. Gloves on, no rubbing eyes, wash hands after.
- **Latex**: poinsettias are in the latex family. Students with a latex allergy wear non-latex (nitrile) gloves or skip handling; check health plans.
- **Bees and pollen**: check IEP/health plans for bee-sting or pollen allergies and know where the EpiPen is before greenhouse work.
- **Mouths**: no plants, soil, seeds, berries, fertilizer or rooting hormone in mouths. Students who mouth objects or have pica need one-to-one eyes in the greenhouse. Many plants (holly and mistletoe berries, some houseplants) are poisonous.
- **Chemicals**: rooting hormone and fertilizer with gloves, then wash hands. Pesticides and growth regulators are staff/teacher only; students never spray or handle them.
- **Greenhouse**: wet floors and hoses (slips), heat (water breaks), wet dry potting mix first (dust).

👩🏾‍🏫 Staff Guide · Leaf Collection Walk (Level 2)

Horticulture 1 · Unit 3 · goes with the class “Leaf Collection” (4 pts). Level 2 = about 3rd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Task card**: Modified from the class “Leaf Collection”. 7 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 3 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- Show me the first step.
- Which tool do we need now?
- Where do our hands go when the tool is cutting?
- What should we do when we finish?

3. Modifications

- Task card instead of written directions
- Sentence starters on the reflection
- Partner reads directions aloud

4. What to grade (4 pts)

- Steps done (photo proof)
- Reflection: 2 sentences per box
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver (SLE).

⚠️ Safety: tools and plants

- **Tools**: students may use all tools (pruners, knives, floral wire, low-temp glue guns) **with an adult right there**. Staff hand tools out, stay at the station, and count them back in. Cut-resistant gloves if possible.
- **Poinsettia sap** is milky and can irritate skin and eyes. Gloves on, no rubbing eyes, wash hands after.
- **Latex**: poinsettias are in the latex family. Students with a latex allergy wear non-latex (nitrile) gloves or skip handling; check health plans.
- **Bees and pollen**: check IEP/health plans for bee-sting or pollen allergies and know where the EpiPen is before greenhouse work.
- **Mouths**: no plants, soil, seeds, berries, fertilizer or rooting hormone in mouths. Students who mouth objects or have pica need one-to-one eyes in the greenhouse. Many plants (holly and mistletoe berries, some houseplants) are poisonous.
- **Chemicals**: rooting hormone and fertilizer with gloves, then wash hands. Pesticides and growth regulators are staff/teacher only; students never spray or handle them.
- **Greenhouse**: wet floors and hoses (slips), heat (water breaks), wet dry potting mix first (dust).

🌿 Parts of the Plant Notes

Horticulture 1 · Unit 3 Parts of the Plant · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Roots and stems

1. One thick main root is a _____; many thin roots are _____.
2. Root _____ absorb water.
3. _____ carries water up; _____ carries food.

Word bank: **fibrous** **hairs** **phloem** **taproot** **Xylem**

B. Leaves

4. The leaf stalk is the _____.
5. Tiny openings for gas exchange are _____.
6. Match:

Form	how leaves sit on stem
Margin	leaf edge
Arrangement	leaf shape

Word bank: **petiole** **stomata**

C. Photosynthesis and flowers

7. Inputs: water and _____.
8. Outputs: food and _____.
9. The male part of the flower is the _____; the female part is the _____.

Word bank: **carbon dioxide** **oxygen** **pistil** **stamen**

Level 2 version of the gen ed “Leaves/Stems/Roots/Flowers worksheets + Unit 3 quiz.”

📖 Parts of the Plant: Vocabulary Picture Cards (Level 2)

Name: _____

Cut on the dashed lines. Use the cards to match, sort, or quiz a partner.



Root

Grows underground, anchors the plant, and takes in water and minerals through root hairs.



Stem

Holds up leaves and flowers and carries water and food through xylem and phloem.



Leaf

The main job is photosynthesis: making food (sugar) from light, water and carbon dioxide.



Flower

The reproductive part. It makes seeds and fruit after pollination.



Photosynthesis

Water + carbon dioxide + light → sugar (food) + oxygen.



Pollination

Moving pollen from the stamen to the pistil, often by bees. It leads to seeds and fruit.



Xylem

Tubes in the stem that carry water and minerals up from the roots.



Phloem

Tubes that carry food (sugar) through the plant.



Stomata

Tiny openings, mostly under the leaf, for gas exchange and transpiration (giving off water).

🌸 Flower Dissection

Horticulture 1 · Unit 3 · Level 2



Name: _____

Modified from the class “Flowers Worksheet / flower dissection”.

🧺 Supplies

- ☐ 🌸 **Flower** lily or alstroemeria
- ☐ 🔪 **Plastic knife or scalpel** adult hands out
- ☐ 🧤 **Gloves**
- ☐ 📄 **Parts mat** label spots
- ☐ 🔍 **Hand lens**
- ☐ 🧻 **Paper towel**

🌸 Steps: check each box when you finish

1. 🧤 Gloves on. Adult hands out the knife and stays at the table. ☐
2. 🌸 Remove sepals and petals; place them on the mat. ☐
3. 🟡 Remove the stamens; find anther and filament. ☐
4. 🟢 Find the pistil: stigma, style, ovary. ☐
5. 🔪 With the adult, cut the ovary across (cut away from you). ☐
6. 🔍 Use a hand lens to find ovules. ☐
7. 📄 Label every part on the mat. ☐
8. 🧼 Return the knife, clean up, wash hands. ☐

💡 *Lily pollen stains clothes and is toxic to cats; alstroemeria is a good low-mess choice.*

🍃 Leaf Collection Walk

Horticulture 1 · Unit 3 · Level 2

Name: _____

Modified from the class “Leaf Collection”.

🧺 Supplies

- ☐ 🛍️ **Collection bag** or tray
- ☐ 🧤 **Gloves** for picking
- ☐ 📋 **Leaf chart** form, margin, arrangement
- ☐ 🔍 **Hand lens** to look close
- ☐ 📸 **Camera/iPad** photos

Steps: check each box when you finish

1.  Gloves on; walk with an adult. ☐
2.  Collect 8 leaves from plants staff approve (no berries, no unknown plants). ☐
3.  Sort by leaf form: needle, round, ovate. ☐
4.  Sort by margin: smooth, dentate, undulate. ☐
5.  Note the arrangement on the stem: opposite, alternate, whorled. ☐
6.  Tape or sketch each leaf and label it. ☐
7.  Wash hands. ☐

🌸 Flower Dissection · Prep Sheet

Horticulture 1 · Unit 3 · Level 2



Name: _____

👥 Job roles: circle your job. Initial when it is done.

Job	What you do	Done ✔️
🧑‍🌾 **Grower**	Does the main plant job with the tools	
🧺 **Supply helper**	Gets and returns supplies and tools to staff	
📸 **Recorder**	Takes photos and fills in the chart	
🧽 **Clean-up captain**	Sweeps, wipes tables, checks hands are washed	

🧰 Tools and supplies: check when you have it

- ☐ 🌸 **Flower** lily or alstroemeria
- ☐ 🔪 **Plastic knife or scalpel** adult hands out
- ☐ 🧤 **Gloves**
- ☐ 📄 **Parts mat** label spots
- ☐ 🔍 **Hand lens**
- ☐ 🧻 **Paper towel**

⚠️ Safety and manners focus

🧤 Wear **gloves** for soil, sap, fertilizer and rooting hormone.

👄 **Nothing in your mouth**: no plants, soil, seeds or berries.

✂️ Sharp tools only with an **adult at the station**. Cut away from your body.

🔪 Knife: **adult at the table**, cut away from you, hand it back when done.

🐝 Pollen allergy? Tell staff; wear a mask or do the picture version.

🧼 **Wash hands** when you finish.

✅ Before you start

☐ Gloves ☐ Apron ☐ Adult at station

🍃 Leaf Collection Walk · Prep Sheet

Horticulture 1 · Unit 3 · Level 2

Name: _____

👥 Job roles: circle your job. Initial when it is done.

Job	What you do	Done ✔️
👩‍🌾 **Grower**	Does the main plant job with the tools	
🧺 **Supply helper**	Gets and returns supplies and tools to staff	
📸 **Recorder**	Takes photos and fills in the chart	
🧽 **Clean-up captain**	Sweeps, wipes tables, checks hands are washed	

🧰 Tools and supplies: check when you have it

- ☐ 🛍️ **Collection bag** or tray
- ☐ 🧤 **Gloves** for picking
- ☐ 📋 **Leaf chart** form, margin, arrangement
- ☐ 🔍 **Hand lens** to look close
- ☐ 📸 **Camera/iPad** photos

⚠️ Safety and manners focus

- 🧤 Wear **gloves** for soil, sap, fertilizer and rooting hormone.
- 👄 **Nothing in your mouth**: no plants, soil, seeds or berries.
- ✂️ Sharp tools only with an **adult at the station**. Cut away from your body.
- ☠️ Only pick leaves an adult approves. **No berries.**
- 🧼 **Wash hands** when you finish.

✅ Before you start

☐ Gloves ☐ Apron ☐ Adult at station

🌸 Flower Dissection · Reflection

Horticulture 1 · Unit 3 · Level 2



Name: _____

📸 My photos / proof

☐ Before photo ☐ During photo ☐ Finished photo

✅ Skills I used today (check all you did)

☐ 🧤 Used gloves and tools safely

☐ 🌱 Finished the Flower Dissection steps

☐ 🧽 Cleaned up and returned tools

☐ 🤝 Did my job role

What did you do today, in order?

1. _____
2. _____

Ideas: **First…** **Next…** **Last…**

What did you learn about plants?

1. _____
2. _____

Ideas: **Plants need…** **I saw…** **Next time…**

🍃 Leaf Collection Walk · Reflection

Horticulture 1 · Unit 3 · Level 2



Name: _____

📸 My photos / proof

☐ Before photo ☐ During photo ☐ Finished photo

✅ Skills I used today (check all you did)

☐ 🧤 Used gloves and tools safely

☐ 🌱 Finished the Leaf Collection Walk steps

☐ 🧽 Cleaned up and returned tools

☐ 🤝 Did my job role

What did you do today, in order?

1. _____
2. _____

Ideas: **First…** **Next…** **Last…**

What did you learn about plants?

1. _____
2. _____

Ideas: **Plants need…** **I saw…** **Next time…**

✅ Parts of the Plant Quiz

Horticulture 1 · Unit 3 · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. 🍃 The stalk that attaches a leaf is the…

A. Pistil | B. Petiole | C. Phloem

2. 🍃 Needle, round and ovate describe…

A. Leaf margins | B. Leaf arrangements | C. Leaf forms

3. 🔁 Opposite, alternate and whorled describe…

A. Leaf arrangements | B. Root types | C. Flower types

4. 💧 Root hairs are responsible for…

A. Making pollen | B. Photosynthesis | C. Water absorption

5. 🌱 The root cap's job is to…

A. Protect the root tip | B. Absorb water | C. Make food

6. 🍬 Phloem moves…

A. Only water upward | B. Food and minerals through the plant | C. Pollen

7. 💧 Xylem moves ___ upward.

A. Sugar | B. Water | C. Seeds

8. ☀️ The inputs of photosynthesis are…

A. Water and carbon dioxide | B. Sugar and oxygen | C. Soil and seeds

9. 🌸 The stamen is…

A. The female part | B. The petal | C. The male part: anther and filament

10. 🍎 Pollination results in…

A. Seed and fruit | B. Roots and stems | C. Leaves

🎬 Silence of the Bees

Horticulture 1 · Unit 3 · Level 2 video guide (modified). PBS Nature: use a short official clip (staff choose).

Name: _____

Before you watch

What do bees do for flowers? _____

While you watch: circle the best answer

1. What do bees carry from flower to flower?

A. Water | B. Pollen | C. Soil

2. Why do farmers need bees?

A. To eat pests | B. To dig soil | C. To pollinate crops so fruit grows

3. What is happening to bee numbers?

A. They are dropping | B. Growing fast | C. Staying the same

4. Which food needs bee pollination?

A. Salt | B. Tap water | C. Apples

After you watch: finish the sentence

5. After watching, finish the sentence:

Bees matter to growers because _____

Goes with the class "Silence of the Bees worksheet" (10 pts). Same points when complete.

🚧 **Under construction:** Full episode upload on YouTube; link a short official PBS clip instead. *Check with the class teacher.*

🌿 Parts of the Plant Notes · ANSWER KEY

Horticulture 1 · Unit 3 Parts of the Plant · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Roots and stems

1. One thick main root is a **taproot**; many thin roots are **fibrous**.
2. Root **hairs** absorb water.
3. **Xylem** carries water up; **phloem** carries food.

Word bank: fibrous · hairs · phloem · taproot · Xylem

B. Leaves

4. The leaf stalk is the **petiole**.
5. Tiny openings for gas exchange are **stomata**.
6. Match:

(Key: Form → leaf shape · Margin → leaf edge · Arrangement → how leaves sit on stem)

Form	how leaves sit on stem
Margin	leaf edge
Arrangement	leaf shape

Word bank: petiole · stomata

C. Photosynthesis and flowers

7. Inputs: water and **carbon dioxide**.
8. Outputs: food and **oxygen**.
9. The male part of the flower is the **stamen**; the female part is the **pistil**.

Word bank: carbon dioxide · oxygen · pistil · stamen

Level 2 version of the gen ed "Leaves/Stems/Roots/Flowers worksheets + Unit 3 quiz."
🔑 Key written by our team from the class teacher's materials, please check.

✅ Parts of the Plant Quiz · ANSWER KEY

Horticulture 1 · Unit 3 · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. 🌿 The stalk that attaches a leaf is the… Q1

A. Pistil | **B. Petiole** | C. Phloem

2. 🌿 Needle, round and ovate describe… Q2

A. Leaf margins | B. Leaf arrangements | **C. Leaf forms**

3. 🔁 Opposite, alternate and whorled describe… Q4

A. Leaf arrangements | B. Root types | C. Flower types

4. 💧 Root hairs are responsible for… Q10

A. Making pollen | B. Photosynthesis | **C. Water absorption**

5. 🌱 The root cap's job is to… Q12

A. Protect the root tip | B. Absorb water | C. Make food

6. 🍬 Phloem moves… Q13

A. Only water upward | **B. Food and minerals through the plant** | C. Pollen

7. 💧 Xylem moves ___ upward. Q25

A. Sugar | **B. Water** | C. Seeds

8. ☀️ The inputs of photosynthesis are… Q27

A. Water and carbon dioxide | B. Sugar and oxygen | C. Soil and seeds

9. 🌸 The stamen is… Q22

A. The female part | B. The petal | **C. The male part: anther and filament**

10. 🍎 Pollination results in… Q23

A. Seed and fruit | B. Roots and stems | C. Leaves

Answers

1-B · 2-C · 3-A · 4-C · 5-A · 6-B · 7-B · 8-A · 9-C · 10-A

Reworded from the gen ed “Unit 3 Parts of the Plant (28 Qs)”. Score: __ / 10. 🔑 Key written by our team from 's materials, please check.

🎬 Silence of the Bees · ANSWER KEY

Horticulture 1 · Unit 3 · Level 2 video guide (modified). PBS Nature: use a short official clip (staff choose).

Name: _____

Before you watch

What do bees do for flowers? _____

While you watch: circle the best answer

1. What do bees carry from flower to flower?

A. Water | **B. Pollen** | C. Soil

2. Why do farmers need bees?

A. To eat pests | B. To dig soil | **C. To pollinate crops so fruit grows**

3. What is happening to bee numbers?

A. They are dropping | B. Growing fast | C. Staying the same

4. Which food needs bee pollination?

A. Salt | B. Tap water | **C. Apples**

After you watch: finish the sentence

5. After watching, finish the sentence:

Bees matter to growers because **they pollinate crops so we get fruit and seeds**

Goes with the class "Silence of the Bees worksheet" (10 pts). Same points when complete.

🚧 **Under construction:** Full episode upload on YouTube; link a short official PBS clip instead. *Check with the class teacher.*

🔑 *Key written by our team from the class teacher's materials, please check.*